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SUMMARY

Senior Principal Researcher focused on **LLM inference efficiency for agentic workloads**. Tech lead and architect at Microsoft M365 Research for end-to-end inference efficiency of agentic workflows: workflow characterization, orchestrator–inference engine co-optimization, KV-cache management, and novel methods for efficient inference.

Recent AI work spans task-specific LLM compression (**LLM-Sieve**, 20–75% parameter reduction at ≤1–5% accuracy loss) and prospection-guided agentic memory (**EMNLP 2026**). Doing ML research since 1997, starting with neural-network pruning and feature selection. Over 20 years at Microsoft Research and Bosch, built widely cited work in networking, IoT and applied ML, and turned research into shipped products, including Azure ML labeling (~10× cost reduction), the next-gen Xbox controller protocol (5× lower latency), and the Orion indoor localization service in Windows Phone.

30+ US patents · 9.5k+ citations · ACM Test of Time Award

CURRENT FOCUS : End-to-End Efficiency for Agentic Workflows.

Agentic workflows comprise multi-step plans, tool calls, loops, many LLM calls per task. Serving them well needs task-level metrics, not just per-request TTFT and TBT, and co-optimization across the stack: prompt and process optimization, model selection, routing and load balancing, batching, and KV-cache retention, eviction and prefetch across turns.

RECENT WORK

Task-Specific LLM Compression (LLM-Sieve)

- Combined evolutionary search with non-orthogonal low-rank projections to prune task-irrelevant capacity: 20–75% parameter reduction at ≤1–5% accuracy loss across domains.

Prospection-Guided Agentic Memory and Personalization (EMNLP 2026, Main Track)

- Introduced Prospection-Guided Retrieval (PGR), inspired by prospection in human cognition: the model imagines the user's likely next steps and uses them as retrieval probes, recovering relevant but semantically distant memories.
- Built MemoryQuest, a public multi-session benchmark for long-horizon personalization.
- Nearly 3× recall over the strongest baseline (1,625 queries, 185 users); responses preferred on 89–98% of queries, confirmed by blinded human annotation.

Technical Skills

- **LLM inference and serving:** KV-cache management, batching and scheduling, orchestrator–engine coordination, workload characterization; familiar with vLLM, LMCache and NVIDIA Dynamo
- **Model efficiency:** pruning, low-rank projection, LoRA, quantization, evolutionary and gradient-based search
- **ML methods:** reinforcement learning, active learning, Bayesian inference, particle filters, evolutionary optimization, human-in-the-loop systems; implemented gradient-descent, genetic-algorithm and Bayesian inference solvers from scratch
- **Systems:** distributed and networked systems, wireless protocols, embedded and real-time systems
- **Languages and tools:** Python, PyTorch, C#, C/C++ (embedded systems)

Selected Impact

- **Azure ML Labeling (Satyam → Azure ML):** crowdsourcing system for ML training-data labeling, with multi-annotation fusion and quality control over non-expert workers; ~10× cost reduction; shipped in Azure ML.
- **Next-gen Xbox controller networking**: co-architected the controller↔console protocol and led two teams (2 researchers, 8 developers); 5× lower latency, 20% lower power, 0.1–0.5% packet loss.
- **On-device ML for gaming hardware** (ADR-X, NSDI 2024): neural-network-assisted Wi-Fi rate adaptation on compute-constrained embedded devices.
- **Cloud gaming sync** (Ekho, SIGCOMM 2023): millisecond-level synchronization of cloud-gaming audio streams across endpoints over the internet; covered by MIT News, Tech Times and ScienceDaily.
- **City-scale IoT incubation:** market sizing to MVP with Compass Group; 300 km² pilot; 30× CAPEX/OPEX reduction; enabled 60% fleet reduction through sharing.
- **Indoor localization** (EZ, Zee, Centaur): ML-driven crowdsourced RF modeling (EZ: hybrid genetic algorithm and gradient descent; Zee: particle-filter Bayesian tracking; Centaur: Bayesian networks) across 50+ venues at 6–10 m accuracy; 2,500+ citations; **shipped in Windows Phone as the Orion localization service**.

EXPERIENCE

Microsoft M365 Research — Senior Principal Researcher · 2026–Present

- Tech lead and architect for end-to-end inference efficiency of agentic workflows in Microsoft 365 Copilot.
- Set technical direction across the stack, from the agent orchestrator to the inference engine: workflow characterization, task-level efficiency metrics, orchestrator–inference engine co-optimization, KV-cache management and novel methods for efficient inference.
- Own the architecture and roadmap, working across research and product engineering teams.

Microsoft Research — Senior Principal Researcher · 2016–2026

- Led applied research in agentic memory, LLM efficiency, human-in-the-loop ML, and networked/IoT systems, with a mandate to turn research into product and set cross-org technical direction.

- Took programs from early problem framing through prototype, pilot, product hardening and transfer: Azure ML labeling, Xbox controller networking, city-scale IoT.
- Built what transfer needs: system architecture, evaluation methodology, failure analysis, operational playbooks and partner onboarding.
- Wrote architecture and decision docs, ran weekly design reviews, and worked with legal on 30+ USPTO filings.
- Mentored researchers, engineers and interns (several now faculty or Staff engineers); led hiring loops.

Microsoft Research India — Senior Researcher · 2009–2016

- Led indoor localization from research to the Orion localization service in Windows Phone, with data collection, calibration and deployment playbooks for 50+ venues.
- Published widely in wireless and embedded systems, working across academia and product teams.

Robert Bosch Research & Technology Center — Senior Research Engineer · 2005–2009

- Delivered wireless industrial control prototypes with 10^{-9} reliability and ≤20 ms latency; owned lab validation and safety cases.

SELECTED PUBLICATIONS

Machine learning

- "Thinking Ahead: Prospection-Guided Retrieval of Memory with Language Models." H. Chopra, K. K. Chintalapudi, S. Nath, R. W. White, C. Shah. *EMNLP 2026 (main track); arXiv 2605.14177.*
- "How Many Parameters Does Your Task Really Need? Task Specific Pruning with LLM-Sieve." W. Reda, A. Jangda, K. Chintalapudi. *arXiv preprint 2505.18350, 2025.*
- "MCAL: Minimum Cost Human-Machine Active Labeling." H. Qiu, K. Chintalapudi, R. Govindan. *ICLR 2023.*
- "A Novel Scheme to Reduce and Determine the Architecture of a Multilayer Perceptron Network." K. K. Chintalapudi, N. R. Pal. *IEEE SMC 1998.*
- "A Simple Procedure to Reduce the Size of a Multilayer Perceptron Network." N. R. Pal, K. K. Chintalapudi. *Int. J. Knowledge Based Engineering Systems, 1998.*
- "Feature Selection in a Connectionist Framework." N. R. Pal, K. K. Chintalapudi. *Int. J. Neural, Parallel and Scientific Computation, 1997.*

Applied machine learning

- "ADR-X: ANN-Assisted Wireless Link Rate Adaptation for Compute-Constrained Embedded Gaming Devices." H. Yin, M. Ramanujam, J. Schaefer, S. Adermann, S. Narlanka, P. Lea, R. Netravali, K. Chintalapudi. *NSDI 2024.*
- "Cloud-LoRa: Enabling Cloud Radio Access LoRa Networks Using Reinforcement Learning Based Bandwidth-Adaptive Compression." M. O. Shahid, D. Koch, J. Raghuram, B. Krishnaswamy, K. Chintalapudi, S. Banerjee. *NSDI 2024.*
- "Kodan: Addressing the Computational Bottleneck in Space." B. Denby, B. Lucia, S. Noghabi, K. Chintalapudi, R. Chandra. *ASPLOS 2023.*
- "Real-Time Video Analytics: The Killer App for Edge Computing." G. Ananthanarayanan, P. Bahl, P. Bodík, K. Chintalapudi, M. Philipose, L. Ravindranath, S. Sinha. *IEEE Computer, 2017.*
- "Enabling Physical Analytics in Retail Stores Using Smart Glasses." S. Rallapalli, A. Ganesan, K. Chintalapudi, V. N. Padmanabhan, L. Qiu. *MobiCom 2014.*
- "Zee: Zero-Effort Crowdsourcing for Indoor Localization." A. Rai, K. K. Chintalapudi, V. N. Padmanabhan, R. Sen. *MobiCom 2012.*
- "Centaur: Locating Devices in an Office Environment." K. Chintalapudi, V. N. Padmanabhan. *MobiCom 2012.*
- "Indoor Localization Without the Pain." K. Chintalapudi, A. Padmanabha Iyer, V. N. Padmanabhan. *MobiCom 2010.*
- "I-MAC: A MAC That Learns." K. Chintalapudi. *IPSN 2010.*

Systems and networking

- "Ekho: Synchronizing Cloud Gaming Media Across Multiple Endpoints." P. Hamadanian, D. Gallatin, M. Alizadeh, K. Chintalapudi. *SIGCOMM 2023.*
- "Concurrent Interference Cancellation: Decoding Multi-Packet Collisions in LoRa." M. O. Shahid, M. Philipose, K. Chintalapudi, S. Banerjee, B. Krishnaswamy. *SIGCOMM 2021.*
- "Mirror in the Sky: Democratizing Continent-Scale IoT Connectivity via Amateur Satellites." J. M. Schellberg, S. Jog, V. Ranganathan, K. Chintalapudi, B. Priyantha, B. Krishnaswamy. *HotMobile 2026.*

AWARDS & SERVICE

ACM Test of Time Award (2022)
FCC TAC — AI Working Group (2022–23)
Reviewer and PC: EMNLP, ICLR, NeurIPS, AISTATS, SIGCOMM, NSDI, MobiCom, MobiSys
Mentored students now faculty at UIUC, UW, Cornell, NYU, Georgia Tech, UC Irvine

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EDUCATION

Exec-Ed, CTO Program — Berkeley Haas, 2025
PhD, Computer Science — USC, 2006
MS, EE (Networking) — USC, 2002
MS, EE (Signal Processing & Control) — Drexel, 1999
BTech, EE — IIT (BHU) Varanasi, 1997